



BB84 Protocol Audit Report

Quantum Key Distribution — Full Session Transcript & Security Analysis

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SECURITY VERDICT

SECURE — KEY ESTABLISHED

EAVESDROPPER (EVE)

✓ NOT DETECTED — CHANNEL SECURE

§1

PROTOCOL CONFIGURATION

BB84 QKD — Simulation Parameters

Protocol	BB84 Quantum Key Distribution (Bennett & Brassard, 1984)
Photon Bases	Rectilinear basis (+): $ 0\rangle, 1\rangle$ Diagonal basis (x): $ +\rangle, -\rangle$
Photons Transmitted (n)	128
Channel Noise Level	9.0% (Noisy Channel)
Eavesdropper Active	No — Channel not compromised by eavesdropper
Error Correction Protocol	LDPC (Belief Propagation) Protocol
Privacy Amplification	Universal Hashing via Toeplitz Matrix Multiplication



INFORMATION-THEORETIC SECRECY AUDIT

Based on the detected **QBER (0.00%)**, the simulator calculated a partial information leakage to Eve of approximately **32.9%**. Privacy Amplification was applied using a **Toeplitz Hash Function** to map the 49-bit working key into a compressed space of **30 bits**.

ENTROPY RETENTION PROFILE



61.2%

STAGE 3 — QUANTUM TRANSMISSION

Alice prepares photons → quantum channel → Bob measures

PHOTONS PREPARED

128

Alice → Quantum Channel

ALICE BITS

128

Raw random bit sequence

ALICE BASES

128

+ and × encodings

BOB MEASUREMENTS

128

Received qubit readings

ALICE RAW BITSTREAM

128 BITS

10111001 01001010 01110111 10100011 00110110 10101010 00000100 11010100 10101001 0
1010101 00101111 11100101 00010011 00101101 10010011 11101001

ALICE BASES (+/×)

128 BASES

××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ×
××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ×××××××

BOB BASES (+/×)

128 BASES

××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ××××××× +
××××××× ××××××× ××××××× ××××××× ××××××× ××××××× ×××××××

BOB MEASURED BITS

128 BITS

01011001 11011001 01001101 10110101 10010011 00000011 00101110 11110100 10101011 0
0010100 00111101 11100000 00111011 00101100 00010010 01001011

Quantum Channel Integrity: No eavesdropper was inserted. Alice transmitted 128 photons over the quantum channel. Bob received and measured each photon independently using a randomly chosen basis (with 9.0% channel noise applied).

STAGE 4 — BASIS SIFTING

Alice and Bob publicly compare bases; keep only matching-basis bits

RAW BITS (INPUT)
128
Pre-sifting

SIFTED KEY LENGTH
65
Basis-matched bits

BITS DISCARDED
63
Basis mismatch

SIFTING RATE
50.8%
~50% expected

● ALICE SIFTED KEY (BITS AT MATCHING-BASIS POSITIONS)

65 BITS

1001101001111000011001000101010001001011111100000110010001001010

● BOB SIFTED KEY (SHOULD MATCH ALICE — ERRORS APPEAR BELOW)

65 BITS

00011010011110000010010001010100010010110111100000110010001001010

Protocol Note: Sifting retains ~50% of raw bits (both parties chose the same basis). In this session, 63 bits were discarded due to basis mismatch, yielding a sifted key of 65 bits. Eve's interception would not be visible at this stage — errors only appear after QBER estimation.

STAGE 5 — QBER ESTIMATION

Quantum Bit Error Rate — Channel Security Parameter Estimation

SIFTED POOL
65
Input bits

BITS SACRIFICED
16
For QBER sample

QBER
0.00%
Threshold: 12.9%

WORKING KEY
49
Post-sacrifice bits

QBER LEVEL

✓ NOT DETECTED — CHANNEL SECURE

0%

12.9% Limit

100%

PARAMETER	VALUE	INTERPRETATION
QBER (Q)	0.00%	Perfect — no detected errors
Security Threshold ($Q_{\max}$)	12.9%	BB84 standard: abort if $Q > 12.9\%$ (Hoeffding Bound)
Bits Sacrificed for QBER	16	~25% of sifted key publicly compared for error detection
Remaining Working Key	49	Proceeds to error correction phase
Eve Detection Confidence	SECURE — No Eve signature	Based on Hoeffding Inequality (statistical bound)

● ALICE WORKING KEY — POST QBER SACRIFICE

49 BITS REMAINING

100100001100010100010011111100000110010001001

INPUT BITS (R_{IN})

49

Post-Reconciliation key

LEAKAGE (I_E)

32.9%

Eve's potential info

PA FACTOR

0.61x

Compression ratio

NET SECRECY

30

Final secure bits (m)

STAGE	PROTOCOL	BITS IN	BITS REMOVED	BITS OUT
Raw Transmission	BB84 Photon Encoding	—	—	128
Basis Sifting	Public basis reconciliation	128	63	65
QBER Sampling	Statistical error estimation	65	16	49
Error Correction	LDPC (Belief Propagation)	49	3	46
PA Leakage Purge	Entropy Nullification ($n \cdot H(Q)$)	46	13	33
PA Safety Margin	Finite-Key Safety Hash	33	3	30

SECURITY CAPACITY BOUND

$$l \leq n - \text{leakage}_{EC} - H_{max}(X|E) \implies l \approx n \cdot (1 - f \cdot H_2(Q))$$

The final key length is mathematically constrained to ensure that even with infinite computing power, the eavesdropper (Eve) possesses $le2^{-k}$ bits of information about the secret key.

Error Correction (LDPC (Belief Propagation)): Alice and Bob reconcile their keys by exchanging parity information over a public classical channel — revealing only whether blocks of bits have even or odd parity, not the bits themselves. This eliminates the 3 bits introduced by channel noise.

Privacy Amplification (Toeplitz Hashing): The key is compressed using a randomly chosen Toeplitz matrix. Any partial knowledge Eve acquired during Error Correction (from public parity broadcasts) is provably eliminated by this compression, resulting in an information-theoretically secure final key.

STAGE 7 — FINAL SECRET KEY

Unconditionally secure shared secret — ready for ChaCha20-Poly1305 encryption

FINAL SECRET KEY — ESTABLISHED QUANTUM-SECURE SYMMETRIC KEY

30 BITS

001110101000001001001001011010

DIGITAL KEY FINGERPRINT (SHA-HASH SIMULATION)

3A82-4968

ENTROPY RETENTION

23.4%

FINAL KEY LENGTH

30

Secure bits

KEY GENERATION EFFICIENCY

23.4%

Final / Raw ratio

PROTOCOL STATUS



SECURE

All security proofs satisfied

RAW

128

Photons



SIFTED

65

Basis Match



WORKING

49

Post-QBER



FINAL

30

Secret Key



Key Ready: Alice and Bob now share an identical, unconditionally secure 30-bit secret key. In practice, this key would be split: half for **ChaCha20** (stream cipher — ensures confidentiality) and half for **Poly1305** (MAC — ensures message authentication and integrity).

SECURITY PARAMETER	VALUE / STATUS	REQUIREMENT	RESULT
QBER	0.00%	$Q \leq 12.9\%$	Pass
Eve Detection	✔ NOT DETECTED — Channel Secure	No eavesdropper	Pass
Key Established	30 bits	> 0 bits	Pass
Key Efficiency	23.4%	Protocol dependent	Normal
Error Correction	LDPC (Belief Propagation)	Cascade / LDPC	✔ Applied
Privacy Amplification	Purge: 13 bits	$H(Q)$ Correction	✔ Leakage Nullified
Key Fingerprint	3A82-4968 ...	Digital Signature	Verified
Overall Verdict	🔒 SECURE — KEY ESTABLISHED		

Academic Context: This report documents a **classical simulation** of the BB84 protocol. All quantum state measurements are performed probabilistically using pseudo-random number generation. Real-world QKD requires photonic hardware including single-photon sources, polarising beam-splitters, and single-photon detectors (SNSPDs). This simulator is designed for *pedagogical demonstration* of quantum cryptographic principles at DIAT's School of Quantum Technology.

